

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

Evaluating AI/ML systems responsibly requires practices for testing, stress-testing, auditing, comparing, and interpreting specific capabilities—not only aggregate benchmark scores. This paper proposes such a practice for spatial reasoning in vision–language models (VLMs): a diagnostic audit of object-relative orientation, which aggregate accuracy can hide. On Visual Spatial Reasoning (VSR), moving from a compact 2.2B model to a 7B VLM substantially improves overall, depth, horizontal, and containment accuracy, but orientation changes only modestly; a large 2026 sparse-MoE model reproduces the pattern (79.5% / 65.7% overall/orientation), a newer flagship (GPT-5.6 Sol) escapes the relative weakness (73.0% orientation), and a third external model (Gemini Spark) yields only a degenerate evaluation (chance-level, TRUE-defaulting; its redo reused the orientation answers, App. A). Except for structured prompting, trained conditions on both controlled models and the sparse-MoE model remain in a narrow 62–66% orientation band. Prompting, LoRA variants, visual-side adaptation, probes, and object-centric decomposition do not reliably close the gap, and a label-sensitivity analysis (an automated audit followed by two blind human re-audits; humans agree well on the failure taxonomy, but exclusion masks remain annotator-dependent and automated auditors vary substantially in calibration) shows that the pattern persists after excluding examples flagged as ambiguous.

Accuracy also fails to capture relational coherence. LM-side adaptation improves consistency over zero-shot; hard-negative training raises facing consistency from 66.0% to 77.7% with facing accuracy unchanged at 68.9% (pooled gain over LM-only n.s.). Cross-dataset evaluation on SITE shows VSR-trained adaptation does not transfer cleanly: neutral overall, significantly harming official spatial-relationship reason-

ing (75.1%→71.6%, $p = 0.004$). A pre-specified orientation-vocabulary slice, frozen before SITE evaluation, is analyzed as exploratory evidence. The results point to an orientation bottleneck that is persistent in the controlled model families and the sparse-MoE model but not universal (GPT-5.6 Sol escapes; Gemini Spark degenerates), in which orientation is not robustly decodable by the tested readouts and relational decisions are not always propagated coherently, while task-specific spatial adaptation can transfer poorly. Each component of the audit—the diagnostic ladder, the label-sensitivity check, the accuracy–coherence separation, and the cross-dataset transfer protocol—is specified operationally so it can be applied to other capabilities, benchmarks, and model families.

1. Introduction

This paper contributes an evaluation practice as much as a set of model results. We develop a protocol for testing, stress-testing, auditing, comparing, and interpreting one capability across vision–language models (VLMs), and instantiate it for object-relative orientation: facing, facing away, parallel, and perpendicular. This capability matters for robotics, embodied agents, navigation, and scene understanding, and is easy to hide behind an aggregate score: identifying the objects, inferring the relation, and answering complementary formulations are separable skills.

Prior benchmarks establish that grounded spatial reasoning remains difficult, but they leave open how to diagnose a relation-specific failure. VSR introduced a controlled natural-image probe and reported particularly poor object-orientation performance [7]; What’sUp and GSR-Bench similarly expose failures on basic grounded relations [6, 9]; and broader evaluations continue to identify orientation as a difficult factor [15]. We therefore ask three narrower questions: does the VSR orientation weakness persist in modern genera-

075 tive VLMs and at much larger scale; can prompting,
076 adaptation, or visual readouts remove it; and do accu-
077 racy, logical coherence, and cross-dataset transfer tell
078 the same story?

079 The novelty is diagnostic rather than a claim that
080 we discovered spatial difficulty. Our contribution is not
081 any individual diagnostic, but a falsification protocol
082 that holds the relation family and examples fixed while
083 testing whether an apparent capability failure survives
084 scaling and adaptation, label re-audit, representation
085 readout, logical reformulation, and cross-dataset trans-
086 fer. Prior work generally reports benchmark accuracy,
087 studies a single intervention, or proposes a consistency
088 objective; one controlled audit of the same family dis-
089 tinguishes a persistent relation-specific bottleneck from
090 aggregate underperformance, annotation artifacts, or
091 benchmark-specific specialization.

092 In our VSR experiments, the distinction between
093 relation families is striking. Moving from SmolVLM2-
094 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-
095 creases zero-shot overall accuracy from 74.0% to 80.9%;
096 horizontal, depth, and containment rise by 6–15 pp.
097 Orientation, by contrast, moves only from 62.8% to
098 63.5%. A large 2026 sparse-MoE model (MiMo-V2.5,
099 310B total / 15B active parameters) reproduces the
100 same relation-specific pattern: 79.5% overall accuracy
101 yet 65.7% on orientation, again inside the 62–66%
102 band. Two further external zero-shot models bound
103 this result: GPT-5.6 Sol does not reproduce the rela-
104 tive weakness—73.0% orientation, above its own depth
105 (67.1%) and outside the band—while Gemini Spark’s
106 evaluation degenerates (51.5% overall; 98.6% TRUE
107 answers in its first pass, a redo that reused the 137 ori-
108 entation verdicts, App. A). The bottleneck is therefore
109 model-family-dependent rather than universal, and one
110 function of the protocol is precisely to detect such
111 differences instead of assuming a failure generalizes.
112 We then evaluate prompting, language-side adaptation,
113 targeted data construction, visual-side adaptation, rep-
114 resentation probing, and object-centric decomposition:
115 structured prompting significantly hurts; general and
116 targeted LoRA improve aggregate accuracy but do not
117 remove the persistent orientation weakness; hard neg-
118 atives improve targeted relational coherence without
119 a significant orientation-accuracy gain; and the tested
120 visual-side LoRA configurations do not outperform the
121 language-side control.

122 As a sensitivity analysis, removing examples flagged
123 by the automated audit raises orientation accuracy
124 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
125 for hard-negative LoRA on the strictest 107-example
126 subset) but does not eliminate error (2B structured
127 shows no gain). The exclusion sets are annotator-

dependent: two blind human re-audits agree strongly
with each other on the failure taxonomy (Krippen-
dorff’s $\alpha=0.81$) while their clean/ambiguous flags agree
only moderately with the automated audit ($\kappa=0.36$ on
the 48 shared persistent-failure cases); details are in the
supplementary. Part of this gain is expected by con-
struction, because the removed examples were drawn
from persistent failures.

A second failure mode emerges from logical consis-
tency. The zero-shot 7B model is consistent on only
36.9% of complementary facing pairs. LM-side LoRA
substantially improves coherence, and hard negatives
raise facing consistency to 77.7% with facing accuracy
unchanged at 68.9%; the extra gain is not significant in
the pooled strict-family comparison, yet accuracy stays
fixed while consistency moves—the metrics capture dif-
ferent behavior.

Finally, we test cross-dataset transfer. The 7B
model is strong on SITE’s official spatial-relationship-
reasoning category (75.1% raw; 59.2% chance-adjusted
accuracy), yet the VSR-trained adapter does not
generalize cleanly: it is statistically neutral overall
(54.2%→53.8%, $p = 0.66$) and significantly degrades
the official category (75.1%→71.6%, $p = 0.004$). A
pre-specified, non-official orientation-vocabulary slice
is much weaker in the aggregate (48.3% raw; 24.0%
chance-adjusted) and shows no positive transfer, but—
as we detail below—its composition limits what it can
establish, and we therefore do not treat SITE as inde-
pendent confirmation of the VSR orientation construct.

Our contributions are four reusable components of
an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR
conditions, two differently sized models, and three
external zero-shot models, several spatial fami-
lies improve substantially while object-relative ori-
entation remains unusually resistant in the con-
trolled families; one newer flagship escapes the re-
lative weakness and one external model’s evaluation
degenerates under the protocol, bounding rather
than weakening the claim. The ladder—scaling,
prompting, language-side adaptation, targeted data
construction, visual-side adaptation, representation
probing, and explicit decomposition—is defined op-
erationally so it transfers to other benchmarks and
relation families.
- A robustness and sensitivity audit. A label-
sensitivity analysis (automated labels with two inde-
pendent human re-audits), frozen/object-grounded
probes, visual-side adaptation, and object-centric
decomposition constrain purely annotation-based or
single-component explanations. The audit practice
(flagged-example exclusion sets, decodability checks

181	against majority baselines) applies to any bench-	232
182	mark whose labels or examples may be ambiguous.	233
183	• An accuracy-coherence separation protocol.	234
184	Complementary-relation tests on VSR's strict	235
185	relation pairs show that, for the tested conditions,	236
186	relational consistency can change substantially	237
187	while task accuracy remains fixed. The protocol	238
188	measures consistency against verdict-pattern base	239
189	rates, so it can be applied wherever complementary	240
190	formulations of the same relation exist.	241
191	• A cross-dataset transfer protocol. SITE shows	242
192	strong official spatial-relationship performance,	243
193	while VSR-trained LoRA transfers poorly and sig-	244
194	nificantly harms the official SITE category; the	245
195	orientation-vocabulary slice is analyzed as ex-	246
196	ploratory evidence. The protocol (pre-specified	247
197	subsets, chance-adjusted accuracy, paired transfer	248
198	tests) is model- and benchmark-agnostic and is re-	249
199	ported with negative as well as positive transfer.	
200	The protocol is deliberately modular: each compo-	
201	nent can be applied independently to other capabilities,	
202	benchmarks, and model families. The supplementary	
203	specifies the prompts, subsets, decision rules, and re-	
204	leased artifacts needed to reproduce the audit.	
205	2. Related Work	
206	Benchmarks for grounded spatial reasoning. VSR in-	
207	troduced a controlled natural-image benchmark with	
208	66 spatial relations and identified facing and parallel	
209	as difficult relations for the VLMs studied at the time [7].	
210	What'sUp controls object identity while varying spa-	
211	tial configuration and reports a substantial human-	
212	model gap [6]; GSR-Bench broadens grounded evalu-	
213	ation across models and scales [9]. SITE extends cov-	
214	erage to single-image, multi-image, and video inputs	
215	and organizes tasks by dimensions such as orientation	
216	and intrinsic/extrinsic relations [15]. SpatialScore and	
217	Mind the Gap further show that spatial competence	
218	remains fragmented [10, 16]. We use VSR not to re-	
219	establish that spatial reasoning is hard, but to turn its	
220	orientation signal into a controlled diagnostic target	
221	and use SITE to test transfer.	
222	Diagnosing and improving spatial failures. Several	
223	lines of work move beyond aggregate accuracy.	
224	AdaptVis relates successful VSR reasoning to attention	
225	on relevant object locations [2], while CREG shows that	
226	attribution evidence in Qwen2-VL can reflect image lay-	
227	out rather than the queried relation [11]. When More Is	
228	Less finds that spatial cues, commonsense information,	
229	and reasoning instructions can help or hurt depend-	
230	ing on the model and setup [1]. Other work probes	
231	visual-concept decodability [13], separates perception	
	from reasoning in spatial planning [17], or supplies ge-	
	ometric inputs for 3D spatial understanding [3]. These	
	motivate our combination of frozen/readout diagnos-	
	tics with adaptation and object-centric decomposition,	
	but none audits one relation family across accuracy,	
	coherence, label sensitivity, and transfer.	
	Taken together, prior work establishes the problem	
	and offers several useful diagnostic lenses, but leaves	
	a methodological gap: benchmark accuracy, represen-	
	tation access, logical consistency, annotation sensitiv-	
	ity, and transfer are usually reported separately. We	
	close that gap with a single controlled diagnostic lad-	
	der, with analysis choices and external-evaluation sub-	
	sets frozen where specified, applied to the same rela-	
	tion family. Our claim is consequently not to discover	
	spatial difficulty, but to make a persistent orientation	
	weakness measurable, falsifiable, and reusable as an	
	evaluation protocol.	
	Adaptation, specialization, and transfer. Parameter-	
	efficient adaptation such as LoRA [5] can improve a	
	benchmark without improving the underlying capa-	
	bility; we therefore compare general, targeted, hard-	
	negative, projector, and vision-plus-projector updates	
	and evaluate the strongest adapter on an untouched	
	benchmark. VSR adaptation is neutral overall on	
	SITE, does not improve the orientation subset, and	
	significantly degrades the official spatial-relationship	
	category—negative transfer treated as an evaluation	
	result rather than an omitted outcome.	
	Representation probing and logical consistency.	
	Probes can test whether a signal is accessible from	
	frozen representations, but probe capacity can itself	
	learn the task, requiring control baselines and cautious	
	interpretation [4]. Consistency-oriented objectives	
	such as SAGE use geometric and linguistic dualities	
	to train spatial reasoning [8]. We instead use probing	
	and complementary-pair consistency as independent	
	diagnostics: the former asks what the tested readouts	
	can decode, while the latter asks whether answers	
	to logically linked statements agree, so accuracy and	
	coherence can diverge as diagnostic axes.	
	3. Experimental Setup	
	3.1. Benchmarks	
	VSR. We use VSR [7] as the primary diagnostic	
	benchmark. The random-split test set contains 2,195	
	examples. The benchmark defines 64 relation labels;	
	61 appear in the evaluated test split. The orientation	
	family contains facing, facing away from, parallel to,	
	and perpendicular to, totaling 137 test examples. All	

281 main VSR results use the same held-out test set and
282 deterministic True/False parsing.

283 SITE. We use SITE [15] for cross-dataset transfer
284 evaluation only and do not train on SITE. The image
285 evaluation contains 2,591 examples: 1,368 single-image
286 and 1,223 multi-image examples. Our primary SITE
287 subset is the official spatial-relationship-reasoning cate-
288 gory ($n = 993$). We additionally report a transparently
289 defined, non-official keyword-derived orientation subset
290 ($n = 2,024$), analyzed as exploratory evidence rather
291 than as direct replication of the VSR construct. SITE
292 results use the benchmark’s native multiple-choice for-
293 mat and chance-adjusted accuracy (CAA) in addition
294 to raw accuracy.

295 3.2. Models and Conditions

296 Our compact model is SmolVLM2-2.2B-Instruct, and
297 our larger controlled-study model is Qwen2-VL-7B-
298 Instruct [14]. We retain Qwen2-VL-7B as the con-
299 trolled anchor because it supports the complete local
300 ladder—the same base model across prompting, LoRA,
301 visual-side adaptation, probing, and decomposition—
302 rather than mixing intervention effects with an uncon-
303 trolled architecture change. We evaluate ten canon-
304 ical VSR conditions: 2B zero-shot, 2B structured
305 prompting, 2B General LoRA, 2B Targeted LoRA,
306 7B zero-shot, 7B General LoRA, 7B Targeted LoRA,
307 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B
308 Vision+Projector LoRA. LoRA [5] is used for all
309 parameter-efficient adaptation conditions. We addi-
310 tionally evaluate a large zero-shot sparse-MoE model,
311 MiMo-V2.5 [12] (310B total / 15B activated param-
312 eters, dedicated 729M-parameter vision encoder, 2026),
313 served through its public API with the identical
314 frozen prompt, thinking disabled, and greedy decoding
315 (App. A). Its role is external validation of the diagnosed
316 pattern; it is not fine-tuned. Two additional external
317 zero-shot models used the same frozen prompt: GPT-
318 5.6 Sol (OpenAI, interactive, 2026-08-12) and Gem-
319 ini Spark (Google, run_gemini_spark_vsr2195_001).
320 Gemini’s first pass defaulted to TRUE on 98.6% of
321 items; its redo reused the 137 orientation answers
322 (id-level verified), so its Table 1 row is a degenerate
323 evaluation finding, not a capability estimate (App. A).

324 General LoRA uses a proportional sample across
325 relation families; Targeted LoRA concentrates train-
326 ing on weak orientation/depth/horizontal families with
327 rehearsal of the rest; Hard-Neg LoRA adds au-
328 dited complementary orientation examples, primarily
329 facing/facing-away inversions; vision-side conditions
330 adapt the multimodal projector or upper vision blocks
331 with it.

332 3.3. Metrics and Statistical Tests

333 For VSR we report accuracy overall and by spatial fam-
334 ily. For SITE we report raw accuracy and CAA. Paired
335 condition comparisons use McNemar tests when ap-
336 plicable. Clean-label orientation analysis reports the
337 full orientation set and progressively stricter audited
338 subsets. Logical consistency is evaluated by construct-
339 ing complementary statements for strict relation pairs
340 (left/right, front/behind, facing/facing-away) and test-
341 ing whether the model assigns opposite verdicts as re-
342 quired. Parallel/perpendicular is treated separately as
343 a soft complement because both relations can be false
344 for oblique objects.

345 4. Main Results

346 4.1. Orientation Remains Weak Across Two Differ- 347 ently Sized VLMs

348 Table 1 shows the central pattern. Moving from the 2B
349 model to the 7B model lifts zero-shot overall accuracy
350 by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and
351 containment by 5.9 pp, but orientation by only 0.7 pp.
352 General 7B LoRA further raises overall accuracy to
353 84.7%, yet orientation reaches only 65.7%. Across
354 trained conditions other than structured prompting,
355 orientation remains in a narrow 62–66% band. The
356 large sparse-MoE zero-shot model (MiMo-V2.5, evalu-
357 ated identically, App. A) falls in the same band: 79.5%
358 overall and 65.7% orientation, with the orientation fam-
359 ily again its weakest relation family (per-family table
360 in the supplementary). The relation-specific pattern is
361 therefore not an artifact of the 2B/7B scale range. Two
362 further external zero-shot models bound the pattern:
363 GPT-5.6 Sol does not reproduce the relative weakness
364 (73.0% vs. its own 67.1% depth; CI [65.0, 79.7] over-
365 laps the band), while Gemini Spark’s evaluation degen-
366 erates (51.5% overall; App. A). The MiMo-V2.5 row
367 is also robust to output-format parsing: orientation
368 is unchanged parseable-only (65.7%), and parseable-
369 only overall is 82.1% vs. 79.5% counting the 69 non-
370 conforming outputs wrong (App. G).

371 A paired bootstrap over the 2,195 shared ex-
372 amples (10,000 resamples) confirms the horizontal-
373 vs-orientation scaling contrast (+13.9 pp; 95% CI
374 [+2.9, +24.8]; positive in 99.3% of resamples); depth
375 (+5.5 pp) and containment (+5.1 pp) are directionally
376 positive but not individually significant (App. D).

377 Structured prompting is a negative result: overall
378 accuracy falls from 74.0% to 68.3% at 2B, while ori-
379 entation falls from 62.8% to 53.3%. A paired exact
380 McNemar test on the full test set confirms that the
381 structured prompt is worse than the minimal baseline
382 ($p \approx 2.2 \times 10^{-7}$). Consistent with recent VSR analy-

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6
GPT-5.6 Sol zero-shot	75.4	73.0	67.1	76.9	82.8	77.8
Gemini Spark zero-shot [†]	51.5	57.7	55.0	53.2	50.3	47.5

[†]Gemini Spark’s evaluation is degenerate (98.6% TRUE in pass 1; redo reused the orientation answers, id-level verified); the row is a degenerate-evaluation finding, not a capability estimate (App. A).

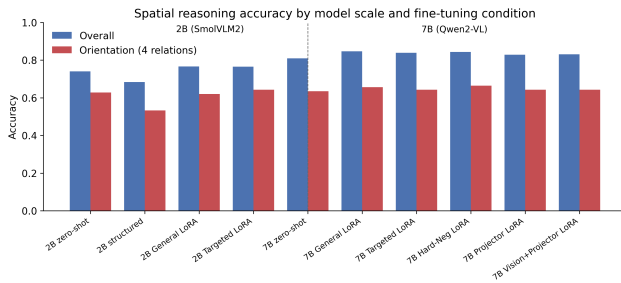


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

ses where additional cues or reasoning instructions help or hurt depending on setup [1], this is specific to the tested 2B condition and prompt.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; frozen v1) flagged ambiguous or questionable orientation examples; we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the automated strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the sample, these are not matched comparisons across families; two blind human re-audits (binary flag on all 137; eight-class taxonomy on the 48 persistent failures) yield three distinct analyses (App. F).

Taxonomy-derived strict masks. Each rater’s strict mask raises orientation accuracy without eliminating error—automated 78.5% ($n=107$), human A 77.6%

($n=107$), human B 78.8% ($n=104$)—and 2B structured prompting still shows no gain under any of the three masks (53.3%, 51.4%, 51.0%).

Binary clean/ambiguous masks. The two humans agree substantially on the 137-example binary flag ($\kappa=0.645$, 82.5% agreement) but only moderately with the automated audit ($\kappa=0.36$ on the 48 cases rated by both sources; the automated audit never flagged the remaining 89), and the taxonomy agrees only weakly with the automated labels ($\alpha=0.14$). Under the human binary masks the gains largely vanish (best 66.7%/66.2% vs. 66.4% full-set), tying gains to taxonomy-derived masks; audit reliability is itself model-dependent—GPT-5.6 Sol ($\kappa=0.34$) and Gemini Spark ($\kappa=0.03$, 126/137 clean) agree with humans only weakly—so LLM-generated ambiguity labels are sensitivity tools, not ground truth (App. F).

Consensus analysis. Under the consensus union mask ($n=62$), trained conditions again improve (7B General LoRA 77.4%) but 7B zero-shot does not (58.1%), and 2B structured’s no-gain pattern does not survive (69.3%), consistent with annotator dependence.

The robust conclusion is narrower: taxonomy-derived strict masks raise orientation scores but do not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise (frozen v1 and both human tables in the supplementary).

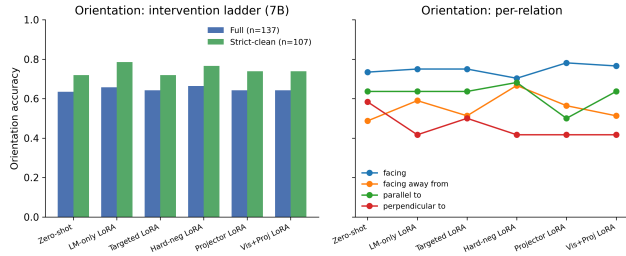


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations (no fine-tuning). Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the CI overlaps the majority baseline.

To reduce the confound that global features infer relations without identifying objects, we extract subject/reference regions and construct grounded features from subject, object, difference, and elementwise-product representations; object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Probing decodability of visual concepts has precedent in driving-oriented VLMs [13] and attribution analyses of this model family [11]; ours targets VSR’s orientation family with pooled, patch-vote, and object-grounded readouts over the model’s exact conditioning features, paired with adaptation and consistency interventions. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, adapting visual or projector components might recover orientation. For the trained checkpoints here it did not: projector and vision+projector LoRA both produced 64.2% orientation, below the 65.7% LM-only

control, with overall nominally lower (82.9%/83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$, not surviving blanket adjustment and not load-bearing, App. D).

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s verdicts match the control’s correctness on only 55.1–58.3% of the 127 boxed statements (per-relation agreement 25.0–69.4%, weakest for parallel); in the committed analysis (10 no-box statements scored wrong) they diverge from the control on 61–65 of 137, with the control correct and the pipeline wrong in 44–47 of them (App. D). These are agreement scores, not ground-truth accuracies: the committed script scored the pipeline against the control’s verdicts, so the decomposition is reported as an agreement diagnostic that fails to reproduce the model’s behavior, not as a direct accuracy comparison (ground-truth scoring via the released script, App. G). Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair consistency has prior literature, including consistency-oriented training [8]; our diagnostic below uses strict pairs with verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The large sparse-MoE model does not repair coherence: 42.7% facing-pair consistency vs. 36.9% (44.0% on the 100 parseable pairs, three excluded for non-conforming outputs), with the same False-defaulting pattern (0 both-True, 56.0% both-False; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The two external models diverge: GPT-5.6 Sol reaches 83.3% complementary facing verdicts (no both-True, $n=102$; 54.5% expected under independence), far above the controlled families, while Gemini Spark’s 60.2% with a zero flip-True rate and 100% complementary front/behind pairs are treated as degenerate, not capability estimates (App. A). The verdict patterns show a False-defaulting

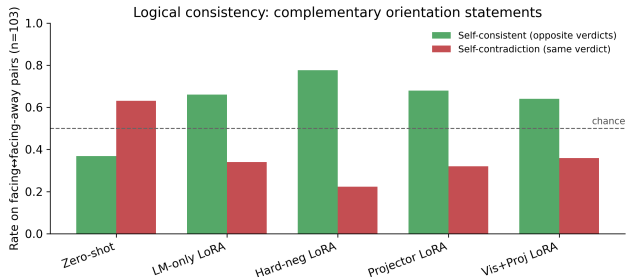


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

bias (both-False 63.1%, both-True 0%, above the 30.2% expected under verdict independence; verdict table in the supplementary).

LM-only LoRA raises facing consistency to 66.0% and front/behind to 70.7% (pooled strict-family significantly better than zero-shot, $p < 0.0001$); hard negatives raise facing consistency to 77.7% with facing accuracy identical at 68.9% (pooled gain over LM-only n.s., $p = 0.29$). The key result is separability: identical facing accuracy coexists with substantially different consistency. Training changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% while both-True contradictions rise from 0% to 24.3%, a disclosed trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2, this adapter run is a post-protocol transfer check rather than a pre-specified step (protocol and outcome in the supplementary). The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified orientation-vocabulary slice is much weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% official) with no positive transfer (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

24% from the spatial-relationship category), we treat it as exploratory evidence rather than replication of the VSR construct. A composition-confound analysis (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and option count, the orientation flag is not significantly associated with correctness (OR 0.84, 95% CI [0.60, 1.16], $p = 0.29$), though descriptively 7.5pp lower within the official category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation [7] while extending it to modern generative VLMs and a broader intervention analysis. The weakness is also model-family-dependent: the

589 controlled families and the sparse-MoE model persist
590 in it, GPT-5.6 Sol does not reproduce it, and Gemini
591 Spark’s evaluation degenerates under the protocol; a
592 diagnostic that only found the failure would be unin-
593 formative, so this distinguishability is what makes the
594 protocol falsifiable.

595 Decodability and coherence both matter, as diagnos-
596 tics. The probes show that orientation is not robustly
597 decodable under the tested frozen-feature readouts,
598 while consistency experiments expose large logical fail-
599 ures even for relations answered above chance. Neither
600 alone identifies a causal mechanism; together they sup-
601 port a mixed account in which orientation information
602 is hard to access under the tested readouts and rela-
603 tional decisions are not always propagated coherently.

604 Task adaptation can specialize rather than generalize.
605 General LoRA improves VSR overall accuracy substan-
606 tially, but its SITE transfer is neutral overall and signif-
607 icantly negative on the official subset. Hard negatives
608 improve a targeted coherence behavior with relation-
609 specific tradeoffs. Benchmark-specific fine-tuning gains
610 should therefore not be equated with general spatial-
611 reasoning improvement.

612 8. Limitations

613 Statistical power is limited at the scale of the orien-
614 tation diagnostic. The VSR orientation family con-
615 tains only 137 test examples, and the smallest per-
616 relation subset, perpendicular, has $n=12$; perpendic-
617 ular results are descriptive only. The Wilson 95% con-
618 fidence intervals on the family-level point estimates are
619 correspondingly wide—62.8% [54, 70], 63.5% [55, 71],
620 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B
621 zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA
622 ($n=137$ each)—so the trained conditions’ 62–66% band
623 should be read as a set of statistically overlapping point
624 estimates rather than a fine-grained ranking. Several
625 smaller point estimates have confidence intervals that
626 include the majority baseline or chance: the patch-
627 vote parallel/perpendicular probe (73.5%, $n=34$; 95%
628 CI [57, 85] vs. the 64.7% majority), the perpendic-
629 ular subset for the trained 7B conditions (41.7%, $n=12$;
630 [19, 68] vs. 50%), and the 7B zero-shot facing-away sub-
631 set (48.7%, $n=39$; [34, 64]). Paired McNemar tests on
632 orientation rest on small discordant-pair counts: the
633 non-significant projector and vision+projector compar-
634 isons ($p=0.86$, $p=0.85$) indicate no detectable effect,
635 not equivalence; conclusions rely on the full-set paired
636 tests and the Holm-adjusted battery (supplementary).

637 The clean-label audit’s exclusion sets originate from
638 an automated LLM audit: two independent human

re-audits reached only moderate per-item agreement 639
with it ($\kappa=0.36$ on the clean/ambiguous flag; Krip- 640
pendorff’s $\alpha=0.14$ on the eight-class taxonomy) while 641
agreeing strongly with each other on the failure taxon- 642
omy ($\alpha=0.81$); both reproduced the qualitative conclu- 643
sions under their own taxonomy-derived strict masks, 644
so the sets are annotator-dependent (tables versioned: 645
frozen v1 plus both human re-audits). The SITE 646
orientation slice is keyword-derived; a composition- 647
confound analysis shows its aggregate score is largely 648
explained by task composition, so SITE is not inde- 649
pendent confirmation of the VSR bottleneck. Our 650
probing methods test accessible decodability under spe- 651
cific linear/MLP and pooling choices; failure to de- 652
code does not prove information is absent. Visual- 653
side experiments use LoRA rather than full retraining; 654
the three model families form a cross-model compar- 655
ison, not a controlled scale experiment. All trained 656
conditions are single-checkpoint runs: paired McNe- 657
mar tests quantify disagreement on the fixed test ex- 658
amples, not training-run variability (a multi-seed ex- 659
tension is prepared in scripts/run_seed_variance.py). 660
The MiMo-V2.5 evaluation is served through a public 661
API: 3.1% of outputs did not conform to the one-word 662
format and are counted wrong (never guessed), and 663
294 contained chain-of-thought text alongside the ver- 664
dict; both are disclosed in the committed predictions 665
(App. A). The MiMo-V2.5 row was evaluated on 2026- 666
08-12 through the OpenCode Go gateway; API-served 667
models can change without notice, so this row’s repro- 668
ducibility depends on the serving snapshot (recorded 669
in results/mimo/). The two additional external rows 670
(App. A) used the identical frozen prompt via third- 671
party interfaces on 2026-08-12; GPT-5.6 Sol’s interac- 672
tive run carries no deterministic-decoding guarantee, 673
and Gemini Spark’s evaluation is degenerate (98.6% 674
TRUE in pass 1; redo reused 1,315/2,195 answers in- 675
cluding all 137 orientation verdicts, id-level verified), 676
so its row and coherence numbers are a documented 677
degenerate-evaluation outcome, not a capability esti- 678
mate. The SITE evaluation is image-only; SITE video 679
reasoning remains future work. Finally, several inter- 680
ventions are evaluated as negative results; lack of im- 681
provement holds for the tested configurations, not as 682
impossibility claims. 683

594 9. Conclusion

Aggregate accuracy hides an orientation bottleneck: 685
adaptation improves several families while leaving ori- 686
entation resistant, a sparse-MoE model reproduces the 687
pattern, a newer flagship escapes it, and one external 688
model degenerates under the protocol. Auditing, prob- 689
ing, and decomposition constrain simple explanations; 690

accuracy and coherence are distinct axes that remain weak in the large model; SITE shows VSR-specific adaptation can negatively transfer. Evaluation should report relation-specific performance, transfer, and logical coherence rather than aggregate accuracy.

References

- [1] Muku Akasaka and Soyeon Caren Han. When more is less: A systematic analysis of spatial and commonsense information for visual spatial reasoning. arXiv preprint arXiv:2602.21619, 2026. 3, 5
- [2] Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. In Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025. 3
- [3] Erik Daxberger, Nina Wenzel, David Griffiths, Haiming Gang, Justin Lazarow, Gefen Kohavi, Kai Kang, Marcin Eichner, Yinfei Yang, Afshin Dehghan, and Peter Grasch. MM-Spatial: Exploring 3d spatial understanding in multimodal LLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 7395–7408, 2025. 3
- [4] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, pages 2733–2743. Association for Computational Linguistics, 2019. 3, 6
- [5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685, 2021. 3, 4
- [6] Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investigating their struggle with spatial reasoning. arXiv preprint arXiv:2310.19785, 2023. 1, 3
- [7] Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. Transactions of the Association for Computational Linguistics, 11:635–651, 2023. 1, 3, 7
- [8] Junming Liu, Yuqi Li, Yifei Sun, Maonan Wang, Piotr Koniusz, Yirong Chen, and Ding Wang. Self-evolving spatial reasoning in vision language models via geometric logic consistency. arXiv preprint arXiv:2605.18162, 2026. 3, 6
- [9] Navid Rajabi and Jana Kosecka. GSR-BENCH: A benchmark for grounded spatial reasoning evaluation via multimodal LLMs. arXiv preprint arXiv:2406.13246, 2024. 1, 3
- [10] Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsafaris. Mind the gap: Benchmarking spatial reasoning in vision-language models. arXiv preprint arXiv:2503.19707, 2025. 3
- [11] Kaizhen Tan, Yang Feng, and Heqing Du. CREG: Compass relational evidence graph for characterizing directional structure in VLM spatial-reasoning attribution. arXiv preprint arXiv:2603.20475, 2026. 3, 6
- [12] Xiaomi MiMo Team. Mimo-v2.5. <https://huggingface.co/collections/XiaomiMiMo/mimo-v25>, 2026. 4
- [13] Nikos Theodoridis, Reenu Mohandas, Ganesh Sistu, Anthony Scanlan, Ciarán Eising, and Tim Brophy. Probing visual concepts in lightweight vision-language models for automated driving. Transactions on Machine Learning Research, 2026. 3, 6
- [14] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. arXiv preprint arXiv:2409.12191, 2024. 4
- [15] Wenqi Wang, Reuben Tan, Pengyue Zhu, Jianwei Yang, Zhengyuan Yang, Lijuan Wang, Andrey Kolobov, Jianfeng Gao, and Boqing Gong. SITE: towards spatial intelligence thorough evaluation. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 9058–9069, 2025. 1, 3, 4, 7
- [16] Haoning Wu, Xiao Huang, Yaohui Chen, Ya Zhang, Yanfeng Wang, and Weidi Xie. Spatialscore: Towards comprehensive evaluation for spatial intelligence. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 31029–31041, 2026. 3
- [17] Qiucheng Wu, Handong Zhao, Michael Saxon, Trung Bui, William Yang Wang, Yang Zhang, and Shiyu Chang. VSP: Diagnosing the dual challenges of perception and reasoning in spatial planning tasks for MLLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 2270–2280, 2025. 3